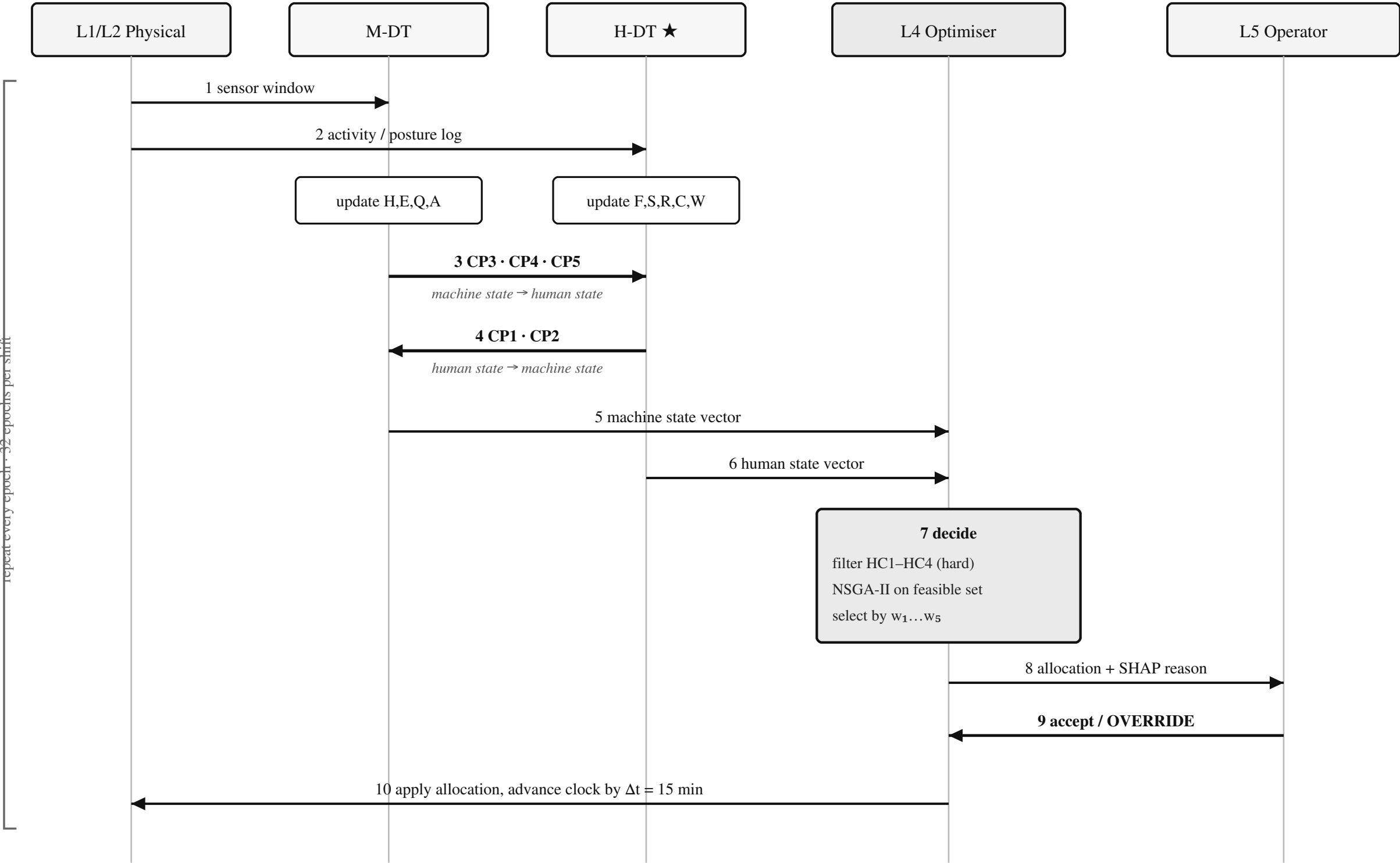


Twin Coupling and Decision Flow within One 15-minute Epoch



Coupling points — the mechanism absent from static formulations

CP1	skill → quality	<i>defect risk rises as skill falls: term $\beta_2(1 - \hat{S}_{h,t})$ in Q_m</i>	CP1, CP2: H-DT → M-DT
CP2	fatigue → quality	<i>a tired operator produces more scrap: term $\beta_3\hat{F}_h$ in Q_m</i>	CP3, CP4, CP5: M-DT → H-DT
CP3	health → cognition	<i>a degraded machine demands attention: term $\gamma_1(1 - H_m)$ in C_h</i>	<i>Removing all five reduces the framework to two independent static parameter sets.</i>
CP4	intensity → fatigue	<i>heavier task raises the fatigue asymptote: $E'_w(\text{task})$ in F_h</i>	
CP5	speed → ergonomics	<i>faster machine worsens posture / safety margin: term $\psi_2\hat{v}_m$ in R_h</i>	

Fig. 2. One decision epoch. Steps 3 and 4 are the bidirectional exchange that distinguishes this framework from static human-aware scheduling.